

FIFA World Cup Match Statistics

Using Team Playing Styles and Machine Learning

1. Introduction

Football is more than a sport-it's a global language, one that brings together people across countries, cultures, and backgrounds. With millions of fans tuning in for major competitions like the FIFA World Cup, it has become one of the most influential and widely celebrated sports on the planet. But beneath the excitement, a deeper question lingers: can we actually understand what makes a team win?

This project sets out to explore that question-specifically, whether after-match statistics can be used to predict which team comes out on top. A team's performance on the pitch is shaped by many interacting factors: possession, passing quality, shot creation, discipline, defensive effort, and physical output, among others. Our goal is to investigate whether these team-level playing-style indicators carry enough signal to classify a team as having won or not won a FIFA World Cup 2026 match.

More precisely, the research question we ask is: **can team-level match statistics-such as possession, passing accuracy, shots on target, fouls, cards, defensive pressure, and distance covered-be used to predict whether a team wins?** Our aim here is twofold: explanatory and predictive. We want to identify which characteristics are most closely associated with winning, while also comparing classification models capable of separating wins from non-wins. It's worth noting that because all features are measured during the match itself, the resulting model should be understood as describing what winning performance looks like, rather than forecasting the outcome of a match that hasn't yet been played.

2. Dataset Description

The dataset was made by collecting data from the FIFA Training Centre Match Report Hub. This website publishes reports after each match, showing team and player statistics. Each row in the dataset is one team from one match, so every match gives us two rows-one for each team. The data used in this project has 176 rows from 88 matches and 48 teams, with 19 original columns. FIFA says these reports cover the main parts of the game, both when a team has the ball and when they don't, and new reports are added as the tournament continues.

Feature group	Variables	Purpose
Ball control	possession_pct; pass_accuracy_pct	Share of possession and completed-pass rate
Attacking output	total_attempts; shots_on_target; corners	Chance creation and attacking pressure
Discipline	fouls; yellow_cards; red_cards	Infractions and sanctions
Work rate/defense	def_pressures; distance_km	Defensive actions and physical coverage
Opponent-relative	possession_diff; pass_accuracy_diff; attempts_diff; corners_diff; pressure_diff	Difference between the team and its opponent
Target	result_label	1 = Won; 0 = Did Not Win

Table
1.

Main variables used in the analysis.

3. Methodology

The project followed the normal steps used in data science: preparing the data, exploring it, engineering features, building models, and checking how well they worked. The dataset had no missing values and no duplicate rows. Text fields were cleaned up, column names were made consistent, and all 15 numerical playing-style features were kept, since none of them were constant in the corrected dataset.

Step 1: Data Preparation

Checked the data carefully: 176 rows, 19 columns, no missing values, and no duplicate rows.

Step 2: Exploratory Analysis

Looked at class frequencies, basic statistics, group averages, distributions, correlations, and a new shot-conversion measure.

Step 3: Leakage-Safe Split

Split the data by match, so both teams from the same match stayed together. Used 70 matches (140 rows) for training and 18 matches (36 rows) for testing.

Step 4: Feature Engineering

Added three new features-shot conversion rate, possession-pass interaction, and discipline index-giving 18 total inputs for the model.

Step 5: Modeling and Tuning

Trained three models (Logistic Regression, Random Forest, and Gradient Boosting) using balanced class weights, GroupKFold cross-validation, GridSearchCV, and threshold tuning.

Step 6: Evaluation

Measured accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices on the test set.

4. Data Preprocessing

Before we could explore the data or build models, we needed to clean and prepare it. The dataset had team-level stats, with each row showing one team's performance in one match.

Loading and Checking the Data

We loaded the corrected dataset into Python using Pandas. It had **176 rows and 19 columns**, covering **88 matches and 48 teams**. Each match appears twice-once for each team.

The dataset had ID-type columns (match_id, team, opponent) plus number-based stats like possession, passing accuracy, shots, corners, fouls, cards, defensive pressure, and distance covered. Team names and match IDs were stored as text, while the performance stats were stored as numbers.

Cleaning Column Names and Team Names

We cleaned up the column names by removing extra spaces and replacing spaces with underscores, giving us names like possession_pct and pass_accuracy_pct. This makes the data easier to work with and avoids mistakes.

We also cleaned the team and opponent columns. For example, different spellings of "Bosnia and Herzegovina" (like "Bosnia & Herz.") were all made consistent, so the same team wasn't accidentally counted as different teams.

Checking for Missing Values

We checked every column for missing values. There were **zero missing values**, so nothing needed to be removed or filled in. We also had a backup plan ready (filling numbers with the median, categories with the most common value) just in case, but since the data was already complete, we didn't need to use it.

Checking for Duplicates

We also checked for duplicate rows and found **zero duplicates**. This matters because duplicate rows could unfairly give some matches more weight in our analysis or model.

Preparing the Target Variable

Our target variable was `result_label`, which shows whether a team won:

- 0 = Did Not Win
- 1 = Won

The data had **120 "Did Not Win" rows and 56 "Won" rows**-about **68%** did not win, and **32%** won. ("Did Not Win" includes both losses and draws, which is why it's the bigger group.) Since the classes were imbalanced, accuracy alone wouldn't tell the full story, so we also used precision, recall, F1-score, confusion matrices, and ROC-AUC.

Choosing Features

We picked 15 number-based features for the model:

`possession_pct`, `pass_accuracy_pct`, `total_attempts`, `shots_on_target`, `corners`, `fouls`, `yellow_cards`, `red_cards`, `def_pressures`, `distance_km`, `possession_diff`, `pass_accuracy_diff`, `attempts_diff`, `corners_diff`, and `pressure_diff`.

These cover different sides of team performance-ball control, passing, attacking, discipline, defense, and physical effort. The "diff" variables compare a team to its opponent (e.g., `possession_diff` is the difference in possession between the two teams). A positive value means the team did better than its opponent; negative means worse.

We left out team names and opponent names on purpose-we wanted the model to learn from playing style and performance, not just learn "this team is historically strong." We also dropped `match_id` since it's just a label, not a real stat.

We double-checked that no feature was the same value for every row. `red_cards`, for example, ranged from 0 to 2, so it had real variation and was kept.

Avoiding Data Leakage

Since each match has two rows (one per team), a normal random split could put one team in training and its opponent in testing-this would leak information between the two. To avoid this, we split the data by **`match_id`**, keeping both teams from the same match together. This gave us **70 matches (140 rows) for training and 18 matches (36 rows) for testing**.

5. Exploratory Data Analysis (EDA)

Next, we explored the data to understand patterns and see which stats might help predict wins. We made a separate copy of the data for this, with the target relabeled as "Won" and "Did Not Win" so it's easier to read.

Basic Statistics

We looked at the mean, standard deviation, min, max, and quartiles for all 15 features.

- Possession ranged from **19.2% to 72.4%** (average: **45.71%**)
- Passing accuracy ranged from **63% to 93%** (average: **84.61%**)
- Total attempts ranged from **2 to 35**; shots on target from **0 to 15**; corners from **1 to 27**
- Defensive pressures ranged from **105 to 471** (average: **256.92**)
- Distance covered was more consistent, averaging **113.90 km** (range: 97.1–160.5 km)

The "diff" variables averaged close to zero, which makes sense—one team's positive difference is balanced by the opponent's negative difference.

Win vs. Loss Counts

A bar chart confirmed the imbalance: 120 non-wins vs. 56 wins. This is why we needed more than just accuracy to judge our models fairly.

Comparing Winners vs. Non-Winners

On average, winning teams had:

- **51.29% possession** vs. 43.10% for non-winners
- **87.29% passing accuracy** vs. 83.36%
- **15.12 total attempts** vs. 11.20
- **5.98 shots on target** vs. 3.46
- **8.02 corners** vs. 6.46

Winning teams also did better relative to their opponents: average possession_diff of **10.70** (vs. -5.00 for non-winners) and average attempts_diff of **5.66** (vs. -2.64).

Discipline, Defense, and Effort

Winning teams committed slightly fewer fouls (**10.75** vs. 11.53) and got fewer yellow cards (**0.91** vs. 1.33). Red cards were rare for both groups (0.04 vs. 0.08), so we're cautious about reading too much into that.

Non-winning teams had more defensive pressures (**266.73** vs. 235.89)—possibly spending more time defending or chasing the ball. Winning teams covered slightly more distance (**115.87 km** vs. 112.98 km), though this difference was small.

Boxplots

We made boxplots for shots on target, passing accuracy, distance, and fouls. Winning teams clearly had higher shots on target and passing accuracy. Distance showed only a small difference, and fouls overlapped a lot between the two groups. So shots on target and passing accuracy separated winners from non-winners better than distance or fouls did.

Attempts vs. Shots on Target

A scatter plot showed that more attempts generally meant more shots on target. Winning teams tended to cluster where both were high; non-winners clustered where both were low. Still, some teams took lots of shots without winning-so shot volume alone doesn't guarantee a win.

Correlation with Winning

The strongest positive correlations with winning were:

- shots_on_target: **0.405**
- attempts_diff: **0.382**
- possession_pct: **0.334**
- possession_diff: **0.323**
- pass_accuracy_pct: **0.309**
- pass_accuracy_diff: **0.303**
- total_attempts: **0.303**

The strongest negative correlations were:

- pressure_diff: **-0.345**
- def_pressures: **-0.213**
- yellow_cards: **-0.189**
- fouls: **-0.092**
- red_cards: **-0.064**

In short: better attacking, possession, and passing were linked to winning, while more defensive pressure and cards were linked to not winning.

A correlation heatmap also showed that some features were related to each other (e.g., possession % and possession_diff, or attempts and shots on target). This means we need to be careful about multicollinearity, especially for Logistic Regression.

Shot Conversion Rate

We created a temporary feature: $\text{shot_conversion_rate} = \text{shots_on_target} \div \text{total_attempts}$. Winning teams converted about **40%** of their attempts into shots on target, compared to **30%** for non-winners. So winners weren't just shooting more-they were more accurate too. This feature was only used for exploration and wasn't included in the final model.

Key Takeaways from EDA

Winning teams generally had stronger attacking play, more shots on target, higher possession, better passing, and better stats relative to their opponent. Non-winning teams faced more defensive pressure and had slightly more fouls and cards.

`shots_on_target` had the strongest single link to winning, but no one variable was strong enough to predict outcomes alone-which is why combining multiple stats through machine learning made sense.

6. Results

Baseline Models

Logistic Regression and Random Forest were trained first, without tuning, on the raw feature set. On the held-out test set, Logistic Regression scored 0.538 F1 and 0.639 ROC-AUC. Random Forest scored slightly lower on F1 (0.522) but higher on ROC-AUC (0.715). These numbers became our baseline for everything that followed.

Cross-Validation

With only 36 rows in the test set, a single evaluation isn't very reliable i.e. the score can shift depending on which matches land in the split. So we used 5-fold GroupKFold cross-validation instead, keeping both rows of a match together in each fold. Both models held up well on ROC-AUC (Logistic Regression 0.783, Random Forest 0.790). Feature engineering clearly helped Random Forest, lifting its F1 from 0.564 to 0.621, but barely moved Logistic Regression (0.564 to 0.567) which is a sign that the new features add non-linear information a tree-based model can use better than a linear one.

Hyperparameter Tuning

Using GridSearchCV, Logistic Regression settled on strong regularization ($C = 0.01$), and Random Forest settled on shallow, constrained trees (depth 5, 100 estimators). Both picked the simpler option which is reasonable, given we only had 140 training rows to work with. On the test set, tuning raised Logistic Regression's F1 to 0.560 and ROC-AUC to 0.719. Random Forest's ROC-AUC rose to 0.733, though its F1 actually dropped to 0.455 which is a reminder that gains from cross-validation don't always carry over perfectly to one small test split.

Gradient Boosting

We added a third algorithm to see if a more flexible model could do better. Tuned Gradient Boosting (learning rate 0.05, depth 3, 50 trees) delivered our strongest single result: F1 0.609, ROC-AUC 0.733.

Threshold Tuning

By default, a prediction above 0.5 counts as a win. Since our target is imbalanced, we tested whether a different cutoff worked better but only using a validation slice from the training data, never the test set, to keep the final evaluation honest. Logistic Regression's best threshold turned out to be the default, 0.5. Random Forest and Gradient Boosting shifted slightly lower (0.47 and 0.31). The resulting F1-scores were close across all three: 0.560, 0.522, and 0.538.

Final Comparison

Across all eight model variants tested, Gradient Boosting (Tuned + Engineered) came out on top where F1 0.609, tied for the best ROC-AUC at 0.733. Logistic Regression followed closely at F1 0.560, and Random Forest placed third on F1 but also tied for the best ROC-AUC.

7. Discussion

Gradient Boosting performed best, but only by a small margin over the much simpler Logistic Regression. That closeness is itself worth noting: Random Forest and Gradient Boosting can both learn complex, non-linear patterns, while Logistic Regression can only draw a straight line through the data. If the real relationship between playing style and winning were highly non-linear, the more flexible models should have pulled well ahead. They actually didn't, which suggests the relationship is largely simple and additive, driven mainly by a few strong features like shots on target.

Each model has a clear trade-off. Logistic Regression is fully transparent i.e. its coefficients are how we found the possession paradox, where possession looks important on its own but fades once shots on target are accounted for. Its downside is that it can't pick up feature interactions unless we build them in ourselves. Random Forest handles those interactions naturally and copes well with the correlated "diff" features from our EDA, but with only 140 training rows it's more prone to overfitting, which is why we constrained its depth. Gradient Boosting gave the best results overall but is the hardest to interpret, since its predictions come from many trees correcting each other rather than a handful of readable weights.

One finding worth flagging directly: our first attempt at threshold tuning picked the cutoff by checking scores on the test set itself which was a leak, since the test set is meant to measure performance, not help choose it. We caught this and fixed it by tuning the threshold on a separate validation slice instead, touching the test set only once, at the end. The corrected scores are a little lower, but honest.

Two limitations matter here. First, with only 88 matches and a 36-row test set, small differences between models carry real sampling noise for which we don't read the ranking as final or definitive. More matches would likely help more than further tuning. Second, and more fundamentally: every feature we use is recorded during the match itself, so this model explains what a winning performance looks like but it doesn't forecast a match that hasn't happened yet, since those inputs simply don't exist beforehand. A genuine forecasting model, say for predicting tournament finishers, would need pre-match team ratings and a bracket simulation instead which is a natural next step for this work.

8. Conclusion

This project successfully developed and evaluated machine learning models to predict FIFA World Cup 2026 match outcomes based on team-level playing style statistics. The primary objective was to determine whether metrics such as possession, passing accuracy, shots on target, and defensive pressure could reliably classify a team as winning or not winning a match.

Exploratory Data Analysis (EDA) revealed significant differences in playing styles between winning and non-winning teams. Winning teams consistently demonstrated superior attacking performance, characterized by higher “shots_on_target”, “total_attempts”, and “corners”. They also exhibited better ball control, evidenced by higher “possession_pct” and “pass_accuracy_pct”, and committed fewer fouls and received fewer cards. Conversely, non-winning teams faced higher defensive pressure. These insights underscored the predictive potential of the selected features.

A critical aspect of this study was the implementation of a leakage-safe group-based splitting strategy using “match_id”. This approach ensured that all data points from a single match were kept together, preventing the artificial inflation of performance metrics that can arise from test-set leakage in time-series or grouped data. This robust methodology provided a more honest and reliable evaluation of model performance.

Following initial baseline models (Logistic Regression and Random Forest), the project incorporated advanced feature engineering. New engineered features, combined with hyperparameter tuning and a group-aware cross-validation framework, significantly improved model performance. Among the models evaluated, the Gradient Boosting Classifier with engineered features emerged as the strongest performer, achieving an F1-score of 0.609 and a ROC-AUC of 0.733 on the hold-out test set. While Logistic Regression offered greater interpretability, Gradient Boosting demonstrated a superior balance of predictive power, suggesting that while relationships are largely additive, capturing some non-linear interactions is beneficial.

The decision to focus solely on the current tournament data was deliberate, driven by several key factors. The FIFA World Cup 2026 marks a significant expansion to a 48-team format, introducing a variety of new participating countries whose historical performance might not accurately reflect their current capabilities. By concentrating on the present tournament, the model emphasizes a team's current form and playing style, rather than relying on past reputation or FIFA rankings. This approach aligns with the project's core philosophy: strong teams don't always win, but teams that consistently create better chances, maintain possession, and defend effectively usually do. Our model is designed to learn from these dynamic patterns of play, offering insights into contemporary football performance.

Despite these advancements, the project acknowledges certain limitations. The dataset, comprising 88 matches and a 36-row test set, is relatively small, meaning model rankings could shift with different data splits. Furthermore, this study focused on predicting match outcomes post-match based on observed statistics, rather than pre-match forecasting. Future work could extend this research by developing a genuine pre-match forecasting model, potentially incorporating team ratings, bracket simulations, and more extensive match data to enhance predictive accuracy and practical applicability.

8. References

1. FIFA. (2026). *FIFA World Cup 2026™ Match Report Hub*.
<https://www.fifatrainingcentre.com/en/game/tournaments/fwc2026/match-reports/>
2. FIFA. (2026). *FIFA World Cup 2026™ Scores & Fixtures*.
<https://www.fifa.com/en/tournaments/mens/worldcup/canadamexicousa2026/scores-fixtures>
3. The Stats Don't Lie. (2026). *2026 FIFA World Cup Statistics*.
<https://www.thestatsdontlie.com/football/world-cup-2026/>